

Yang Zheng

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Education

New York University, Tandon School of Engineering | M.S. in Computer Science *Expected May 2027*
Relevant Coursework: Cloud Computing, Big Data, Software Engineering, Computer Networking, Applied Cryptography, Blockchain & DLT

New York University, College of Arts & Science | B.A. in CS & Mathematics *2021 - 2025*

Experience

Research Assistant | Robotics and Automation Laboratory, Tsinghua University *June – Aug 2023*

- Developed a data-driven method to estimate 3-D states of deformable linear objects from single-frame occluded point clouds, addressing real-world occlusion challenges in robotic perception.
- Trained a two-branch neural network exploiting global and local point cloud information; designed a fusion module combining both approaches for improved 3-D state estimation accuracy.

Projects

Dining Concierge Chatbot | Python, JavaScript *Spring 2026*
AWS S3, API Gateway, Lambda, Lex, SQS, DynamoDB, OpenSearch, SES, Yelp Fusion API

- Delivered personalized restaurant recommendations from 1,300+ Manhattan restaurants by architecting a serverless pipeline across 9 AWS services (Lex, Lambda, API Gateway, SQS, DynamoDB, OpenSearch, SES, S3).
- Improved user retention by implementing DynamoDB-backed state persistence across 5 conversational intents, enabling returning users to retrieve and reuse previous dining preferences.

Productivity Tools and Weekly Planner | React, Node.js, MongoDB, Bootstrap *Spring 2023*

- Built a full-stack web application implementing productivity workflows inspired by "The 7 Habits of Highly Effective People," featuring weekly planning, task prioritization, and habit tracking.

Programming Language Interpreter | Scala *Spring 2025*

- Implemented a full interpreter for a subset of JavaScript including lexing, parsing, and evaluation, supporting closures, higher-order functions, and recursive data structures.

Graphics Animations and Programs | JavaScript, WebGL, OpenGL *Fall 2024*

- Developed ray tracing and rasterization renderers; created real-time animations of human motion, ocean waves, clouds, texture mapping, and a solar system simulation.

Technical Skills

Languages: C++, Python, Java, JavaScript, C, SQL, R, Scala, HTML/CSS

Frameworks & Tools: React, Node.js, MongoDB, PyTorch, WebGL, OpenGL, Bootstrap, Git

Cloud & Infrastructure: AWS (S3, Lambda, API Gateway, Lex, SQS, SES, DynamoDB, OpenSearch), Serverless Architecture